1. Virtual Network Setup

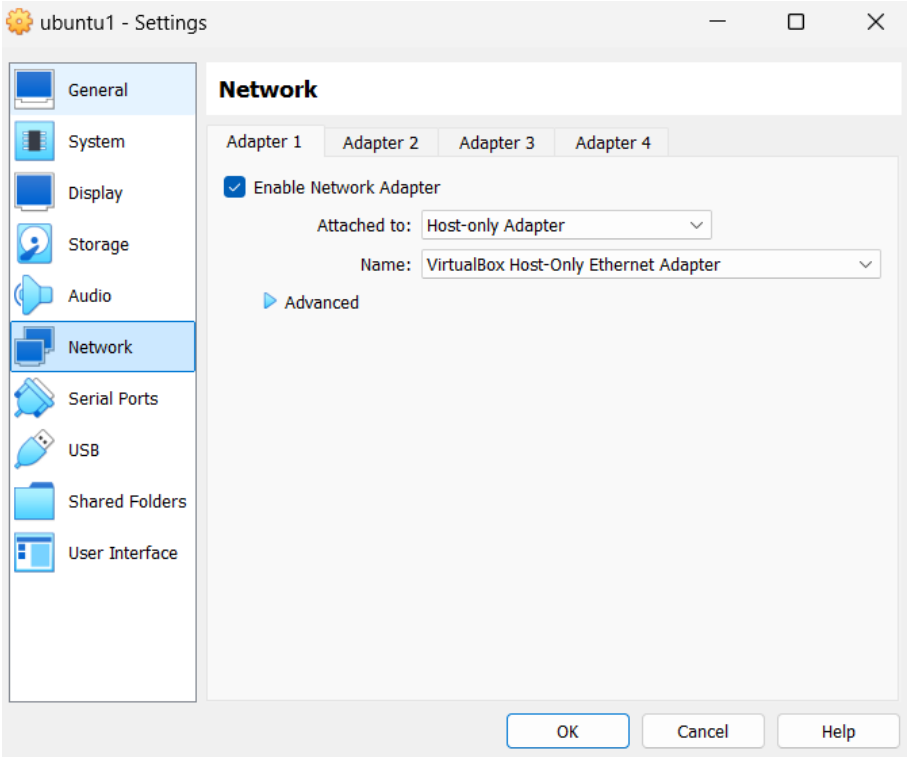


Figure (1) ubuntu-1 device Verify the host-only network

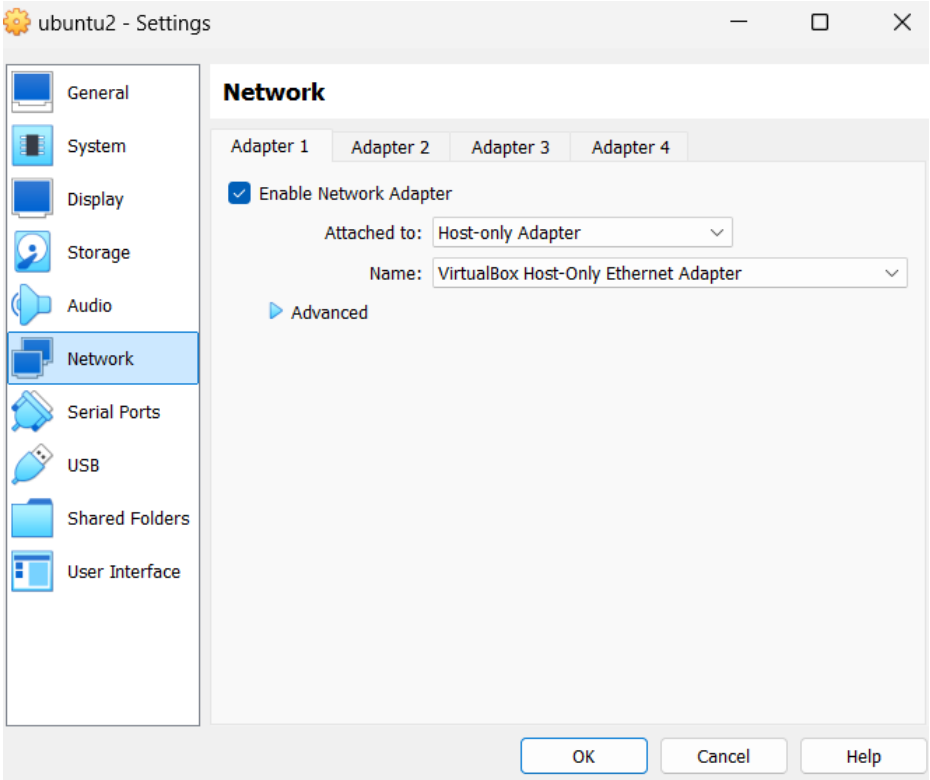


Figure (2) ubuntu-2 device Verify the host-only network

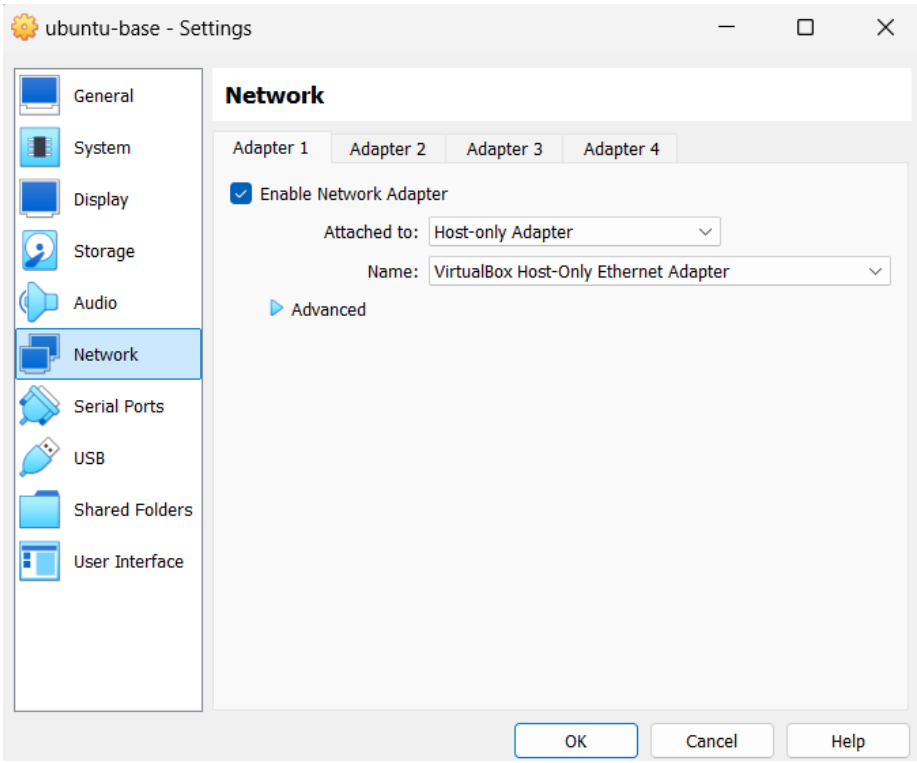


Figure (3) ubuntu-base device Verify the host-only network

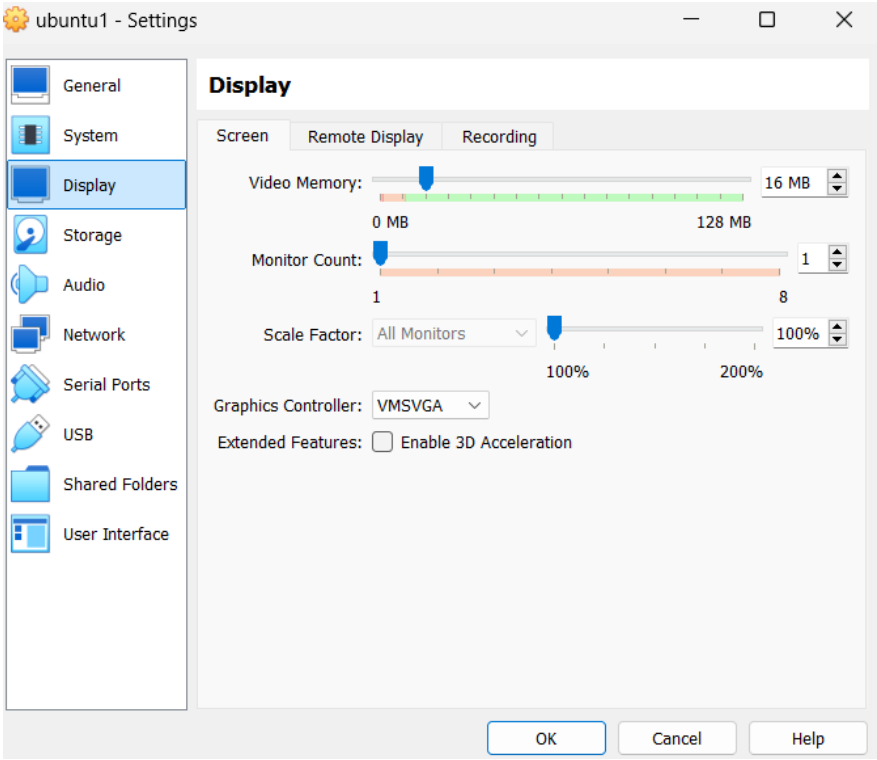


Figure (4) ubuntu-1 device Verify the Proper configuration

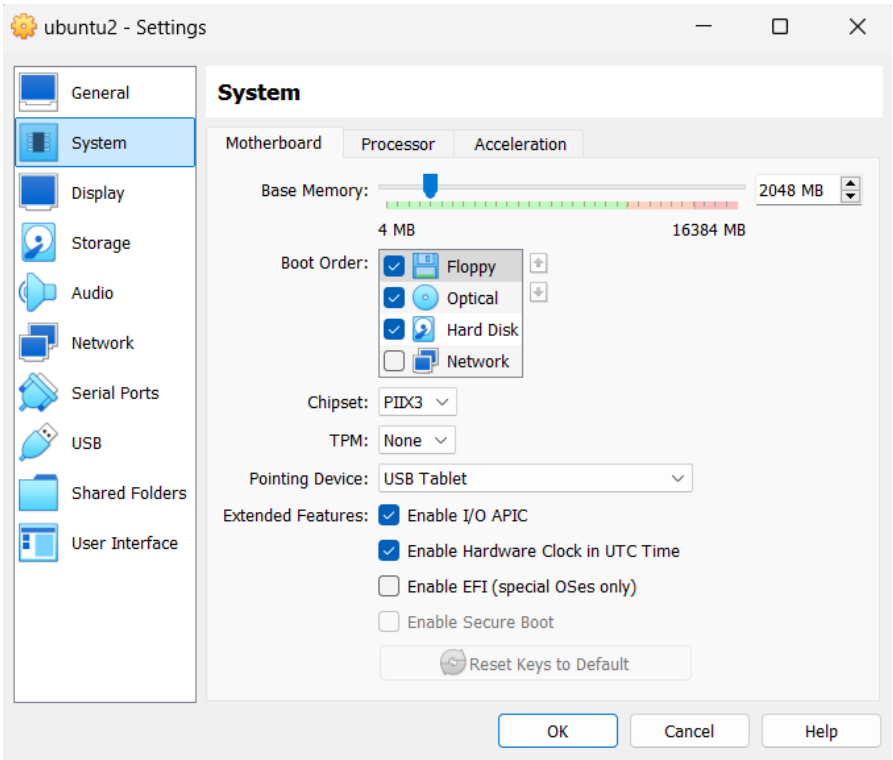


Figure (5) ubuntu-2 device Verify the Proper configuration

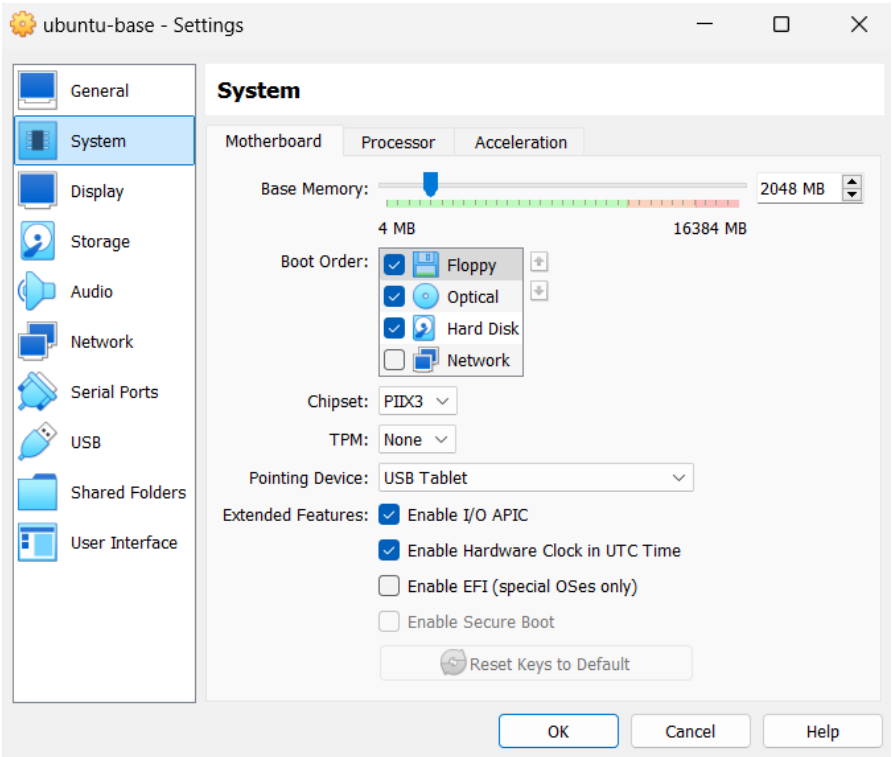


Figure (6) ubuntu-base device Verify the Proper configuration

2. Verify the Network Configuration

1. ip addr Command

```
ubuntu@ubuntu-base:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau
lt qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:54:68:15 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global noprefixroute enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe54:6815/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:60:37:55 brd ff:ff:ff:ff:ff:ff
4: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:1e:95:6a brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.8/24 brd 192.168.56.255 scope global dynamic noprefixroute
enp0s9
       valid_lft 341sec preferred_lft 341sec
    inet6 fe80::4074:87d3:4b6:8a9/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
ubuntu@ubuntu-base:~$
```

Figure (7) ubuntu-base device Verify the Network Configuration

```
ubuntu@ubuntu-1:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau
lt qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:b8:b6:bb brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global noprefixroute enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feb8:b6bb/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:66:79:10 brd ff:ff:ff:ff:ff:ff
4: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:82:8e:3f brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.4/24 brd 192.168.56.255 scope global dynamic noprefixroute
enp0s9
       valid_lft 352sec preferred_lft 352sec
    inet6 fe80::2d66:963b:192e:51f2/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
ubuntu@ubuntu-1:~$
```

Figure (8) ubuntu-1 device Verify the Network Configuration

```
ubuntu-2@ubuntu-2:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau
lt qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
                                                        ff:ff
    inet 192.168.56.103/24 brd 192.168.56.255 sc  e global noprefixroute enp0s3
    inet6 fe80::a00:27ff:fe26:28e1/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:11:0f:2f brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.5/24 brd 192.168.56.255 scope global dynamic noprefixroute
enp0s8
       valid_lft 382sec preferred_lft 382sec
    inet6 fe80::9ee:8b57:b90d:1b2a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
ubuntu-2@ubuntu-2:~$
```

Figure (9) ubuntu-2 device Verify the Network Configuration

- **Procedure:**
  The ip addr command was executed on each VM to check the network settings.
- **Results:**
  1- The first machine (Ubuntu Base): has an IP address of 192.168.56.101.
  2- The second machine (ubuntu-1): has an IP address of 192.168.56.102.
  3- The third machine (ubuntu-2): has an IP address of 192.168.56.103.
  All network interfaces are in UP mode.

3. Ping Other Machines

```
ubuntu@ubuntu-base: ~
ubuntu@ubuntu-base:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=1.75 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.770 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=1.05 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.848 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=0.782 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=1.38 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=1.17 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=0.750 ms
```

Figure (10) ubuntu-base device Ping Verify with ubuntu-1

```
ubuntu@ubuntu-base: ~
ubuntu@ubuntu-base:~$ ping 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.668 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.738 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.743 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.538 ms
64 bytes from 192.168.56.103: icmp_seq=5 ttl=64 time=0.830 ms
64 bytes from 192.168.56.103: icmp_seq=6 ttl=64 time=0.825 ms
64 bytes from 192.168.56.103: icmp_seq=7 ttl=64 time=0.783 ms
64 bytes from 192.168.56.103: icmp_seq=8 ttl=64 time=0.631 ms
64 bytes from 192.168.56.103: icmp_seq=9 ttl=64 time=0.757 ms
64 bytes from 192.168.56.103: icmp_seq=10 ttl=64 time=0.760 ms
```

Figure (11) ubuntu-base device Ping Verify with ubuntu-2

```
ubuntu-2@ubuntu-2:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.474 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.745 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.527 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.795 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=0.845 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=0.505 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=0.710 ms
```

Figure (12) ubuntu-2 device Ping Verify with ubuntu-base

```
ubuntu-2@ubuntu-2:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=1.03 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.786 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.775 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.926 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=0.461 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=0.314 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=0.453 ms
```

Figure (13) ubuntu-2 device Ping Verify with ubuntu-1

```
ubuntu@ubuntu-1:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.620 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.606 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.874 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.01 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=0.625 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=0.660 ms
```

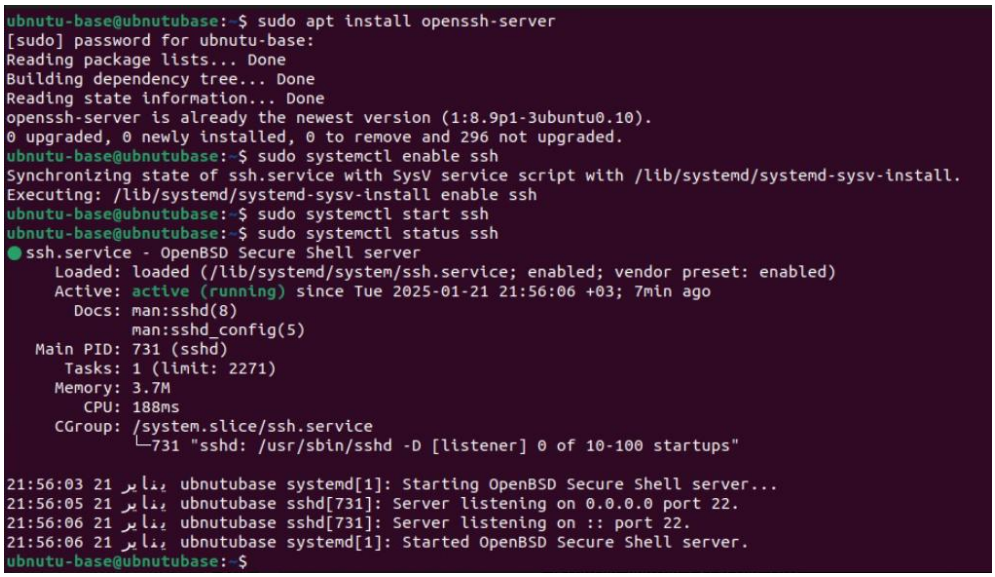
Figure (14) ubuntu-1 device Ping Verify with ubuntu-base

```
ubuntu@ubuntu-1:~$ ping 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.593 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.806 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.566 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.922 ms
64 bytes from 192.168.56.103: icmp_seq=5 ttl=64 time=0.863 ms
64 bytes from 192.168.56.103: icmp_seq=6 ttl=64 time=0.786 ms
64 bytes from 192.168.56.103: icmp_seq=7 ttl=64 time=0.470 ms
64 bytes from 192.168.56.103: icmp_seq=8 ttl=64 time=0.387 ms
```

Figure (15) ubuntu-1 device Ping Verify with ubuntu-2

- **Procedure:**
  Ping commands were executed from the Ubuntu Base machine to the ubuntu-1 and ubuntu-2 machines and from the ubuntu-1 machine to the ubuntu-base and ubuntu-2 and from the ubuntu-2 machine to the ubuntu-base and ubuntu-1.
- **Results:**
  Connection is successful for all : all packets are sent and received successfully, with an average response time of less than 1 ms without loss, with similar response time.

4. Establish SSH Connections



```
ubnutu-base@ubnutubase:~$ sudo apt install openssh-server
[sudo] password for ubnutu-base:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:8.9p1-3ubuntu0.10).
0 upgraded, 0 newly installed, 0 to remove and 296 not upgraded.
ubnutu-base@ubnutubase:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
ubnutu-base@ubnutubase:~$ sudo systemctl start ssh
ubnutu-base@ubnutubase:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2025-01-21 21:56:06 +03; 7min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 731 (sshd)
      Tasks: 1 (limit: 2271)
     Memory: 3.7M
        CPU: 188ms
     CGroup: /system.slice/ssh.service
             └─731 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

21:56:03 21 يناير ubnutubase systemd[1]: Starting OpenBSD Secure Shell server...
21:56:05 21 يناير ubnutubase sshd[731]: Server listening on 0.0.0.0 port 22.
21:56:06 21 يناير ubnutubase sshd[731]: Server listening on :: port 22.
21:56:06 21 يناير ubnutubase systemd[1]: Started OpenBSD Secure Shell server.
ubnutu-base@ubnutubase:~$
```

Figure (16) Correct installation, configuration, and enabling of the SSH service on ubuntu-base

```
ubuntu1@ubuntu1-VirtualBox:~$ sudo apt install openssh-server
[sudo] password for ubuntu1:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:8.9p1-3ubuntu0.10).
0 upgraded, 0 newly installed, 0 to remove and 296 not upgraded.
ubuntu1@ubuntu1-VirtualBox:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
ubuntu1@ubuntu1-VirtualBox:~$ sudo systemctl start ssh
ubuntu1@ubuntu1-VirtualBox:~$ sudo systemctl stautus ssh
Unknown command verb stautus.
ubuntu1@ubuntu1-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: e>
     Active: active (running) since Tue 2025-01-21 22:10:42 +03; 5min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 740 (sshd)
      Tasks: 1 (limit: 2271)
     Memory: 3.8M
        CPU: 166ms
     CGroup: /system.slice/ssh.service
             └─740 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

22:10:40 21 يناير ubuntu1-VirtualBox systemd[1]: Starting OpenBSD Secure Shell >
22:10:42 21 يناير ubuntu1-VirtualBox sshd[740]: Server listening on 0.0.0.0 por>
22:10:42 21 يناير ubuntu1-VirtualBox systemd[1]: Started OpenBSD Secure Shell s>
22:10:42 21 يناير ubuntu1-VirtualBox sshd[740]: Server listening on :: port 22.
lines 1-16/16 (END)...skipping...
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2025-01-21 22:10:42 +03; 5min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 740 (sshd)
      Tasks: 1 (limit: 2271)
     Memory: 3.8M
        CPU: 166ms
```

Figure (17) Correct installation, configuration, and enabling of the SSH service on ubuntu-1

```
ubuntu2@ubuntu2-VirtualBox:~$ sudo apt install openssh-server
[sudo] password for ubuntu2:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:8.9p1-3ubuntu0.10).
0 upgraded, 0 newly installed, 0 to remove and 296 not upgraded.
ubuntu2@ubuntu2-VirtualBox:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
ubuntu2@ubuntu2-VirtualBox:~$ sudo systemctl start ssh
ubuntu2@ubuntu2-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: e>
     Active: active (running) since Tue 2025-01-21 22:22:28 +03; 5min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 730 (sshd)
      Tasks: 1 (limit: 2271)
     Memory: 3.7M
        CPU: 179ms
     CGroup: /system.slice/ssh.service
             └─730 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

22:22:26 21 يناير ubuntu2-VirtualBox systemd[1]: Starting OpenBSD Secure Shell >
22:22:28 21 يناير ubuntu2-VirtualBox sshd[730]: Server listening on 0.0.0.0 por>
22:22:28 21 يناير ubuntu2-VirtualBox sshd[730]: Server listening on :: port 22.
22:22:28 21 يناير ubuntu2-VirtualBox systemd[1]: Started OpenBSD Secure Shell s>
lines 1-16/16 (END)...skipping...
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
     Active: active (running) since Tue 2025-01-21 22:22:28 +03; 5min ago
```

Figure (18) Correct installation, configuration, and enabling of the SSH service on ubuntu-2

```
ubuntu@ubuntu-base:~$ ssh ubuntu@192.168.56.102
ubuntu@192.168.56.102's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

630 updates can be applied immediately.
427 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan  9 10:29:42 2025 from 192.168.56.101
ubuntu@ubuntu-1:~$
```

Figure (19) SSH connect From Ubuntu-Base, to ubuntu-1

```
ubuntu@ubuntu-base:~$ ssh ubuntu-2@192.168.56.103
ubuntu-2@192.168.56.103's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

630 updates can be applied immediately.
427 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu-2@ubuntu-2:~$
```

Figure (20) SSH connect From Ubuntu-Base, to ubuntu-2

```
ubuntu@ubuntu-1:~$ ssh ubuntu@192.168.56.101
ubuntu@192.168.56.101's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

607 updates can be applied immediately.
395 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan  9 11:53:39 2025 from 192.168.56.103
ubuntu@ubuntu-base:~$
```

Figure (21) SSH connect From Ubuntu-1 to ubuntu- Base

```
ubuntu@ubuntu-1:~$ ssh ubuntu-2@192.168.56.103
ubuntu-2@192.168.56.103's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

630 updates can be applied immediately.
427 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Jan 10 03:39:55 2025 from 192.168.56.102
ubuntu-2@ubuntu-2:~$
```

Figure (22) SSH connect From Ubuntu-1 to ubuntu-2

```
ubuntu-2@ubuntu-2:~$ ssh ubuntu@192.168.56.101
ubuntu@192.168.56.101's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

607 updates can be applied immediately.
395 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan  9 11:37:42 2025 from 192.168.56.102
ubuntu@ubuntu-base:~$
```

Figure (23) SSH connect From Ubuntu-2 to ubuntu- Base

```
ubuntu-2@ubuntu-2:~$ ssh ubuntu@192.168.56.102
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established
.
ED25519 key fingerprint is SHA256:AvtglfDv2oDw9+uRFwjVq24r/1Q+Tx3qGgtfh/a9MHE.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.102' (ED25519) to the list of known host
s.
ubuntu@192.168.56.102's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

630 updates can be applied immediately.
427 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan  9 10:34:55 2025 from 192.168.56.101
ubuntu@ubuntu-1:~$
```

Figure (24) SSH connect From Ubuntu-2 to ubuntu-1

- **Procedure:**

Used ssh commands to connect from Ubuntu Base to the other two machines and from ubuntu-1 to the other two machines and from ubuntu-2 to the other two machines.

- **Results:**
  Connection between Ubuntu-Base and ubuntu-1 and ubuntu-2 was successful after entering the credentials.
  Connection between ubuntu-1 and ubuntu-2 and Ubuntu-Base was also successful.
  Connection between ubuntu-2 and ubuntu-1 and Ubuntu-Base was also successful.

5. Create a Text File on ubuntu-1 via SSH

1. From Ubuntu Base, SSH into ubuntu-1.

2. using the following command:

```
echo "This is a test file from Ubuntu Base." > testfile.txt
```

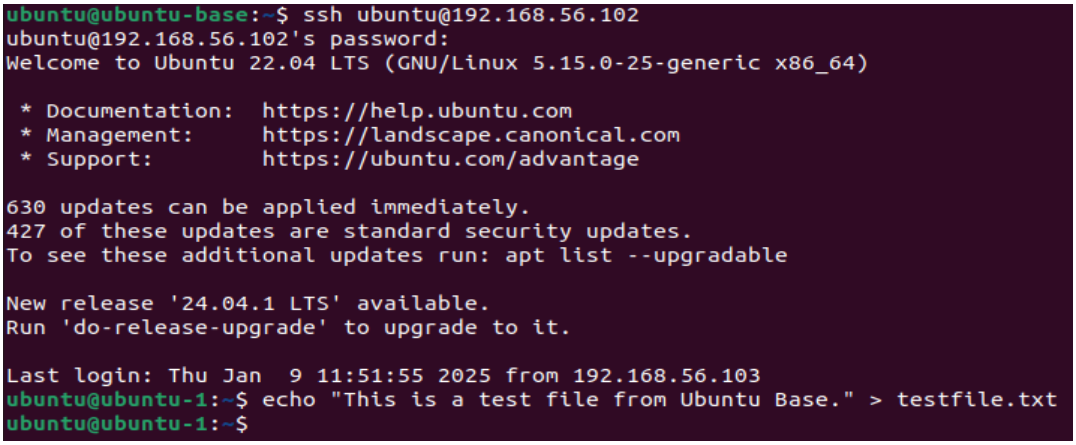


Figure (25) SSH Create a text file From Ubuntu-bass to ubuntu-1

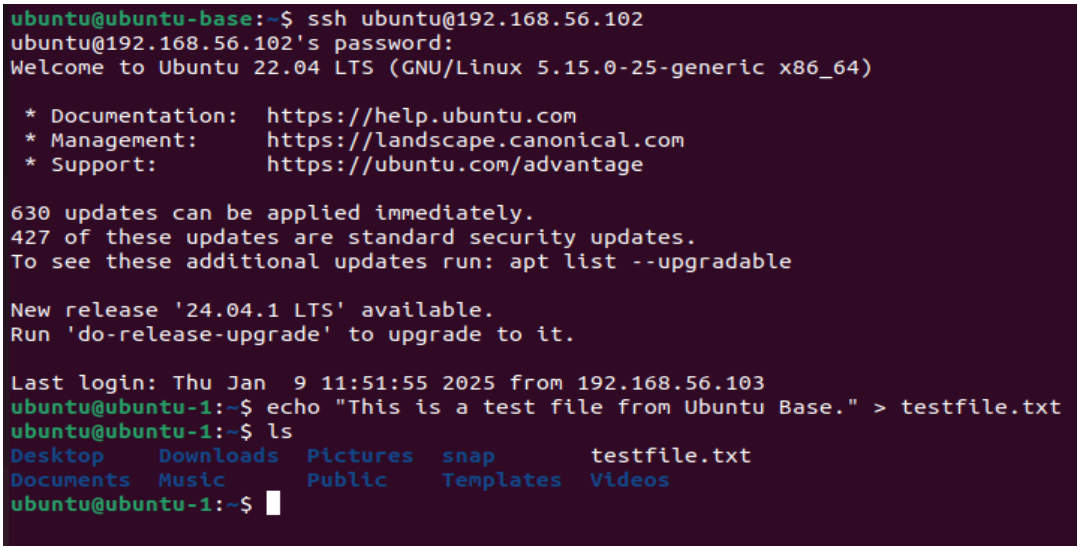


Figure (26) Verify the file was created by listing the directory contents

Logged into ubuntu-1 via SSH and Created a text file using the command:

echo "This is a test file from Ubuntu Base." > testfile.txt

Verified the file creation using ls.

- **Results:**
  The file testfile.txt was created successfully and contains the specified text.
  The file appeared under the contents of the current directory when verified using ls.

6. Transfer a File to ubuntu-2 via SCP

```
ubuntu@ubuntu-base:~$ scp testfile.txt ubuntu-2@192.168.56.103:/home/ubuntu-2
ubuntu-2@192.168.56.103's password:
testfile.txt                                  100%   38    47.6KB/s   00:00
ubuntu@ubuntu-base:~$
```

Figure (27) From Ubuntu Base, transfer the file testfile.txt to ubuntu-2 using SCP

```
ubuntu-2@ubuntu-2:~$ ls /home/ubuntu-2
Desktop    Downloads  Pictures  snap       testfile.txt
Documents  Music      Public    Templates  Videos
ubuntu-2@ubuntu-2:~$
```

Figure (28) Verify the file transfer by SSHing into ubuntu-2

- **Procedure:**
  The file testfile.txt was moved from Ubuntu-Base to the ubuntu-2 machine using the Command
  scp testfile.txt ubuntu-2 @192.168.56.103:/home/ubuntu-2
- **Results:**
  The file was successfully moved.
  When checking for the file on ubuntu-2 using ls /home/ubuntu-2 , the file appeared in the specified location.

7. Restart the SSH Service on ubuntu-1

1. From Ubuntu Base, SSH into ubuntu-1.

```
ubuntu@ubuntu-base:~$ ssh ubuntu@192.168.56.102
ubuntu@192.168.56.102's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

630 updates can be applied immediately.
427 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan  9 12:07:41 2025 from 192.168.56.101
ubuntu@ubuntu-1:~$ sudo systemctl restart ssh
[sudo] password for ubuntu:
ubuntu@ubuntu-1:~$
```

Figure (29) From Ubuntu Base, SSH into ubuntu-1 Restart the SSH service

```
ubuntu@ubuntu-base:~$ ssh ubuntu@192.168.56.102
ubuntu@192.168.56.102's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

630 updates can be applied immediately.
427 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan  9 12:07:41 2025 from 192.168.56.101
ubuntu@ubuntu-1:~$ sudo systemctl restart ssh
[sudo] password for ubuntu:
ubuntu@ubuntu-1:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: >
     Active: active (running) since Thu 2025-01-09 12:12:36 EST; 16min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 21549 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCE>
   Main PID: 21550 (sshd)
      Tasks: 1 (limit: 2292)
     Memory: 1.7M
        CPU: 17ms
```

Figure (30) Verify the service is running

- **Procedure:**
  Logged into ubuntu-1 via SSH.
  Restarted the SSH service using: sudo systemctl restart ssh
  Checked the service status using: sudo systemctl status ssh
- **Results:**
  The service was restarted successfully.
  The service status as shown is:
  Active: running
  No errors are visible.

8. Configure Firewall on All Machines

1- Firewall enabled on all machines using sudo ufw enable, SSH traffic was verified to be allowed using: sudo ufw allow ssh.

```
ubnutu-base@ubnutubase:~$ sudo apt install ufw
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.1-4ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 296 not upgraded.
ubnutu-base@ubnutubase:~$ sudo ufw allow ssh
Skipping adding existing rule
Skipping adding existing rule (v6)
ubnutu-base@ubnutubase:~$ sudo ufw allow from any to any
Skipping adding existing rule
Skipping adding existing rule (v6)
ubnutu-base@ubnutubase:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubnutu-base@ubnutubase:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
Anywhere                   ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
Anywhere (v6)              ALLOW       Anywhere (v6)

ubnutu-base@ubnutubase:~$
```

Figure (31) Installation and configuration of ufw with rules for allowing SSH and ICMP traffic on ubuntu-base

```
ubuntu1@ubuntu1-VirtualBox:~$ sudo apt install ufw
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.1-4ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 296 not upgraded.
ubuntu1@ubuntu1-VirtualBox:~$ sudo ufw allow ssh
Skipping adding existing rule
Skipping adding existing rule (v6)
ubuntu1@ubuntu1-VirtualBox:~$ sudo ufw allow from any to any
Skipping adding existing rule
Skipping adding existing rule (v6)
ubuntu1@ubuntu1-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu1@ubuntu1-VirtualBox:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
Anywhere                   ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
Anywhere (v6)              ALLOW       Anywhere (v6)

ubuntu1@ubuntu1-VirtualBox:~$
```

Figure (32) Installation and configuration of ufw with rules for allowing SSH and ICMP traffic on ubuntu-1

```
ubuntu2@ubuntu2-VirtualBox:~$ sudo apt install ufw
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.1-4ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 296 not upgraded.
ubuntu2@ubuntu2-VirtualBox:~$ sudo ufw allow ssh
Skipping adding existing rule
Skipping adding existing rule (v6)
ubuntu2@ubuntu2-VirtualBox:~$ sudo allow from any to any
sudo: allow: command not found
ubuntu2@ubuntu2-VirtualBox:~$ sudo ufw allow from any to any
Skipping adding existing rule
Skipping adding existing rule (v6)
ubuntu2@ubuntu2-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu2@ubuntu2-VirtualBox:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
Anywhere                   ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
Anywhere (v6)              ALLOW       Anywhere (v6)

ubuntu2@ubuntu2-VirtualBox:~$
```

Figure (33) Installation and configuration of ufw with rules for allowing SSH and ICMP traffic on ubuntu-2

```
ubuntu@ubuntu-1:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

Figure (34) Verify Firewall on ubuntu-1

```
ubuntu@ubuntu-base:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

Figure (35) Verify Firewall on ubuntu-base

```
ubuntu-2@ubuntu-2:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)

ubuntu-2@ubuntu-2:~$
```

Figure (36) Verify Firewall on ubuntu-2

2- Attempted to connect via SSH between machines:

From Ubuntu Base to ubuntu-1:

From Ubuntu Base to ubuntu-2:

From ubuntu-1 to ubuntu-2:

Results:

All SSH connections were successful after enabling the firewall.

The credentials were entered, and the login was successful without any issues.

```
ubuntu@ubuntu-base:~$ ssh ubuntu@192.168.56.102
ubuntu@192.168.56.102's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

630 updates can be applied immediately.
427 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan  9 13:10:11 2025 from 192.168.56.103
```

Figure (37) SSH connections From Ubuntu-Base to ubuntu-1

```
ubuntu@ubuntu-base:~$ ssh ubuntu-2@192.168.56.103
ubuntu-2@192.168.56.103's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

630 updates can be applied immediately.
427 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Jan 10 05:09:27 2025 from 192.168.56.102
ubuntu-2@ubuntu-2:~$
```

Figure (38) SSH connections From Ubuntu-base to ubuntu-2

```
ubuntu@ubuntu-1:~$ ssh ubuntu@192.168.56.101
ubuntu@192.168.56.101's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

607 updates can be applied immediately.
395 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan  9 13:05:22 2025 from 192.168.56.103
```

Figure (39) SSH connections From Ubuntu-1 to ubuntu-base

```
ubuntu@ubuntu-1:~$ ssh ubuntu-2@192.168.56.103
ubuntu-2@192.168.56.103's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

630 updates can be applied immediately.
427 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Jan 10 04:09:42 2025 from 192.168.56.101
ubuntu-2@ubuntu-2:~$
```

Figure (40) SSH connections From Ubuntu-1 to ubuntu-2

```
ubuntu-2@ubuntu-2:~$ ssh ubuntu@192.168.56.102
ubuntu@192.168.56.102's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

630 updates can be applied immediately.
427 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan  9 12:12:25 2025 from 192.168.56.101
ubuntu@ubuntu-1:~$
```

Figure (41) SSH connections From Ubuntu-2 to ubuntu-1

```
ubuntu-2@ubuntu-2:~$ ssh ubuntu@192.168.56.101
ubuntu@192.168.56.101's password:
Welcome to Ubuntu 22.04 LTS (GNU/Linux 5.15.0-25-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

607 updates can be applied immediately.
395 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jan  9 12:58:07 2025 from 192.168.56.101
ubuntu@ubuntu-base:~$
```

Figure (42) SSH connections From Ubuntu-2 to ubuntu-base

3- ICMP (Ping) traffic was allowed using: sudo ufw allow proto icmp

Ping commands were executed to test connectivity between devices:

From Ubuntu Base to ubuntu-1:ping 192.168.56.102

From Ubuntu Base to ubuntu-2:ping 192.168.56.103

From ubuntu-1 to ubuntu-2:ping 192.168.56.103

Results:

All connectivity tests (Ping) succeeded:

Packets were sent and received without loss.

Latency within expected limits.

```
ubuntu@ubuntu-base:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.339 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.821 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.620 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.721 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=0.474 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=0.855 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=1.25 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=0.992 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=64 time=0.816 ms
```

Figure (43) Ping from Ubuntu-base to ubuntu-1

```
ubuntu@ubuntu-1:~$ ping 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.818 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.956 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=1.05 ms
```

Figure (44) Ping from Ubuntu-1 to ubuntu-2

```
ubuntu@ubuntu-1:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.933 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.695 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.607 ms
```

Figure (45) Ping from Ubuntu-1 to ubuntu-base

```
ubuntu-2@ubuntu-2:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.679 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.724 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.669 ms
```

Figure (46) Ping from Ubuntu-2 to ubuntu-base

```
ubuntu-2@ubuntu-2:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.568 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.952 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=1.04 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.825 ms
```

Figure (47) Ping from Ubuntu-2 to ubuntu-1

Results:

Active rules included:

Allow SSH traffic.

Allow ICMP traffic.

Case Study: Monitoring Network Traffic with Wireshark

Analyze Traffic in Wireshark

1. **dentify IP addresses involved in the connections**

   Procedure:

   Traffic was monitored using Wireshark after applying the filters.

   The following addresses were identified during the connections:

   Ubuntu Base: 192.168.56.101

   ubuntu-1: 192.168.56.102

   ubuntu-2: 192.168.56.103

   Results:

   The above IP addresses appeared as either source or destination depending on the connection type.

   The flow to and from the devices was clearly identified in Wireshark.

ip.addr == 192.168.56.101

No.	Time	Source	Destination
5	7.530645734	192.168.56.101	8.8.8.8
6	7.546894403	192.168.56.101	8.8.8.8
7	7.677557813	8.8.8.8	192.168.56.101
8	7.771949520	8.8.8.8	192.168.56.101
12	10.735578610	192.168.56.101	224.0.0.251
13	27.736033617	192.168.56.101	185.125.190.56
14	42.986375886	192.168.56.101	185.125.190.56
15	44.656151169	185.125.190.56	192.168.56.101
20	71.073545224	192.168.56.101	8.8.8.8
21	71.085835497	192.168.56.101	8.8.8.8
22	71.086197008	192.168.56.101	8.8.8.8
23	71.508850741	8.8.8.8	192.168.56.101
24	71.691720030	8.8.8.8	192.168.56.101
25	71.696665202	8.8.8.8	192.168.56.101
26	73.295711627	192.168.56.101	8.8.8.8

Figure (48) Filter traffic by ip address

ip.addr == 192.168.56.102

No.	Time	Source
656	98.501808018	192.168.56.102
657	98.722490800	8.8.8.8
658	98.722766215	192.168.56.102
659	98.722766363	192.168.56.102
660	98.722945491	192.168.56.102
661	98.722945613	8.8.8.8
662	98.857615637	8.8.8.8
663	98.857766540	192.168.56.102
664	98.866074923	8.8.8.8
665	98.866242392	192.168.56.102
682	107.098455662	192.168.56.102
683	107.363819106	91.189.91.48
684	107.364138122	192.168.56.102
685	107.364283628	192.168.56.102
686	107.398832118	91.189.91.48
687	107.604467549	91.189.91.48
688	107.604467929	91.189.91.48
689	107.604711151	192.168.56.102
690	107.604974830	192.168.56.102
691	107.604974973	91.189.91.48
694	108.863657301	192.168.56.102
695	108.863725559	8.8.8.8
696	108.929607707	8.8.8.8
697	108.930199397	192.168.56.102
718	118.030147475	192.168.56.102
719	118.192195361	185.125.190.58
874	169.195491393	192.168.56.102
875	169.304878595	8.8.8.8

Ethernet II, Src: PcsCompu_b8:b6:bb (08:

Figure (49) Filter traffic by ip address

ip.addr == 192.168.56.103

No.	Time	Source
1444	346.189129398	192.168.56.103
1451	346.205314042	192.168.56.103
1452	346.214493302	192.168.56.103
1453	346.214493568	192.168.56.103
1454	346.263486891	192.168.56.103
1455	346.317707503	8.8.8.8
1456	346.318522039	192.168.56.103
1457	346.324424368	8.8.8.8
1458	346.329536849	8.8.8.8
1464	346.446160117	192.168.56.103
1467	346.635087921	192.168.56.103
1468	346.641721596	8.8.8.8
1469	346.641722447	91.189.91.49
1471	346.641811387	192.168.56.103
1473	346.641867594	192.168.56.103
1474	346.642234178	192.168.56.103
1477	346.683482107	91.189.91.49
1478	346.699783691	192.168.56.103
1480	346.733525216	8.8.8.8
1481	346.736161962	192.168.56.103
1484	346.882141337	91.189.91.49
1485	346.882141755	91.189.91.49
1486	346.882196292	192.168.56.103
1487	346.882528517	192.168.56.103
1488	346.882568899	91.189.91.49
1498	347.184393133	8.8.8.8
1499	347.189560585	192.168.56.103
1500	347.189742463	192.168.56.103

Figure (50) Filter traffic by ip address

ip.dst == 192.168.56.103

ip.dst_host

No.		Source	Destination
1455	346.317707503	8.8.8.8	192.168.56.103
1457	346.324424368	8.8.8.8	192.168.56.103
1458	346.329536849	8.8.8.8	192.168.56.103
1468	346.641721596	8.8.8.8	192.168.56.103
1469	346.641722447	91.189.91.49	192.168.56.103
1477	346.683482107	91.189.91.49	192.168.56.103
1480	346.733525216	8.8.8.8	192.168.56.103
1484	346.882141337	91.189.91.49	192.168.56.103
1485	346.882141755	91.189.91.49	192.168.56.103
1488	346.882568899	91.189.91.49	192.168.56.103
1498	347.184393133	8.8.8.8	192.168.56.103
1504	347.276108367	8.8.8.8	192.168.56.103
1507	347.363223961	8.8.8.8	192.168.56.103
1603	357.608520323	8.8.8.8	192.168.56.103
1604	357.817160658	8.8.8.8	192.168.56.103
1666	375.603514061	8.8.8.8	192.168.56.103
1667	375.614281450	8.8.8.8	192.168.56.103
1669	375.784899495	185.125.190.57	192.168.56.103
1674	376.323207937	8.8.8.8	192.168.56.103
1675	376.327769372	8.8.8.8	192.168.56.103
1687	381.399366903	8.8.4.4	192.168.56.103
1689	381.580299908	8.8.4.4	192.168.56.103
1716	408.334186971	8.8.4.4	192.168.56.103
1717	408.341275990	185.125.190.57	192.168.56.103
1724	408.905266975	192.168.56.101	192.168.56.103
1727	408.905603979	192.168.56.101	192.168.56.103
1728	408.905898492	192.168.56.101	192.168.56.103
1731	408.913846457	192.168.56.101	192.168.56.103

Figure (51) Filter traffic by destination ip address

ip.dst == 192.168.56.102

No.	Time	Source	Destination
657	98.722490800	8.8.8.8	192.168.56.102
661	98.722945613	8.8.8.8	192.168.56.102
662	98.857615637	8.8.8.8	192.168.56.102
664	98.866074923	8.8.8.8	192.168.56.102
683	107.363819106	91.189.91.48	192.168.56.102
686	107.398832118	91.189.91.48	192.168.56.102
687	107.604467549	91.189.91.48	192.168.56.102
688	107.604467929	91.189.91.48	192.168.56.102
691	107.604974973	91.189.91.48	192.168.56.102
695	108.863725559	8.8.8.8	192.168.56.102

Figure (52) Filter traffic by destination ip address

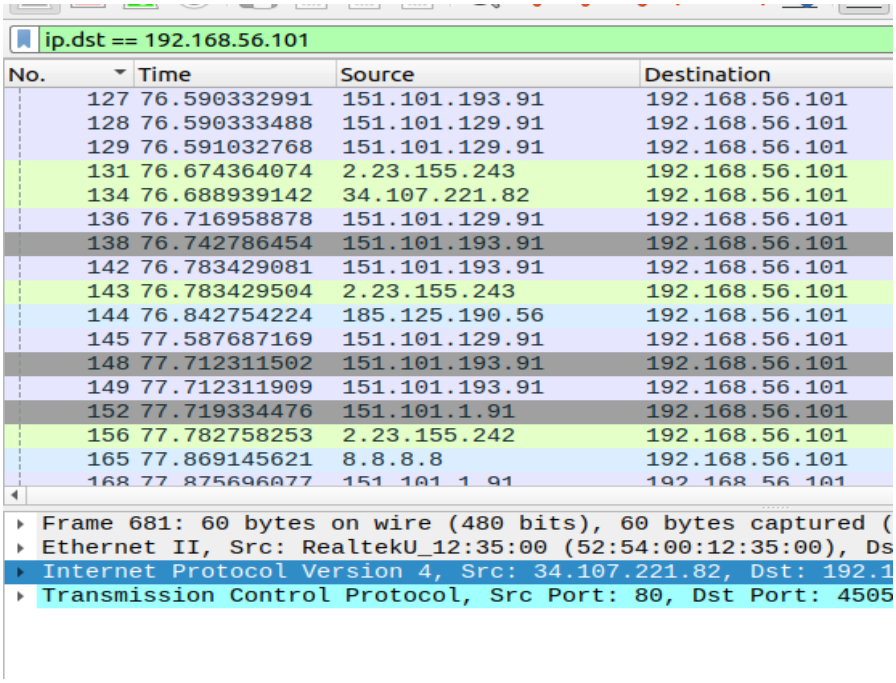


Figure (53) Filter traffic by destination ip address

2. Filter Traffic by Protocol

Procedure:

Wireshark was started and the appropriate network interface (enp0s3) was selected to capture packets.

The following filters were used to analyze the different protocols:

SSH:To analyze SSH traffic.

ICMP: To analyze Ping traffic.

TCP:To analyze general TCP traffic.

Results:

SSH: Displays encrypted traffic generated by SSH connections between devices.

ICMP: Appears ICMP packets that express successful Pings between devices.

TCP: Displays general TCP connections (file transfers using SCP).

ssh

No.	Time	Source	Destination	Protocol	Length	Info
987	196.291466718	192.168.56.101	192.168.56.102	SSHv2	108	Client: Protocol (SSH-2.0
989	196.297967698	192.168.56.102	192.168.56.101	SSHv2	108	Server: Protocol (SSH-2.0
991	196.299042364	192.168.56.102	192.168.56.101	SSHv2	1178	Server: Key Exchange Init
993	196.302486326	192.168.56.101	192.168.56.102	SSHv2	1602	Client: Key Exchange Init
995	196.303793440	192.168.56.101	192.168.56.102	SSHv2	114	Client: Elliptic Curve Di
997	196.307840645	192.168.56.102	192.168.56.101	SSHv2	590	Server: Elliptic Curve Di
998	196.317189329	192.168.56.101	192.168.56.102	SSHv2	82	Client: New Keys
1000	196.346888223	192.168.56.101	192.168.56.102	SSHv2	110	Client: Encrypted packet
1002	196.347219023	192.168.56.102	192.168.56.101	SSHv2	110	Server: Encrypted packet
1003	196.347259388	192.168.56.101	192.168.56.102	SSHv2	134	Client: Encrypted packet
1004	196.354186900	192.168.56.102	192.168.56.101	SSHv2	118	Server: Encrypted packet
1053	201.367559896	192.168.56.101	192.168.56.102	SSHv2	150	Client: Encrypted packet
1055	201.417598485	192.168.56.102	192.168.56.101	SSHv2	94	Server: Encrypted packet
1057	201.417888752	192.168.56.101	192.168.56.102	SSHv2	178	Client: Encrypted packet
1059	201.546610580	192.168.56.102	192.168.56.101	SSHv2	694	Server: Encrypted packet

Figure (54) Filter traffic by Protocol (SSH)

icmp

No.	Time	Source	Destination	Protocol	Length	Info
1120	235.616592036	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) request id=0
1121	235.617220378	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) reply id=0
1124	236.629231410	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) request id=0
1125	236.629801611	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) reply id=0
1128	237.653855135	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) request id=0
1129	237.654425623	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) reply id=0
1130	238.678407938	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) request id=0
1131	238.678881709	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) reply id=0
1134	239.701293971	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) request id=0
1135	239.701883190	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) reply id=0
1136	240.725274086	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) request id=0
1137	240.725844642	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) reply id=0
1140	241.749443226	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) request id=0
1141	241.749956320	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) reply id=0
1142	242.773091793	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) request id=0

```
▸ Frame 1120: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface enp0s3, id 0
▸ Ethernet II, Src: PcsCompu_54:68:15 (08:00:27:54:68:15), Dst: PcsCompu_b8:b6:bb (08:00:27:b8:b6:bb)
▸ Internet Protocol Version 4, Src: 192.168.56.101, Dst: 192.168.56.102
▸ Internet Control Message Protocol
```

Figure (55) Filter traffic by Protocol (ICMP)

tcp

No.	Time	Source	Destination	Protocol	Length	Info
34	74.034422208	192.168.56.101	34.107.221.82	TCP	74	45056 → 80 [SYN] Seq=0 Wi
38	74.241407474	192.168.56.101	34.117.188.166	TCP	74	47298 → 443 [SYN] Seq=0 W
39	74.289060176	192.168.56.101	34.107.221.82	TCP	74	45058 → 80 [SYN] Seq=0 Wi
40	74.496715409	192.168.56.101	34.117.188.166	TCP	74	47308 → 443 [SYN] Seq=0 W
41	74.608822884	34.117.188.166	192.168.56.101	TCP	60	443 → 47298 [SYN, ACK] Se
42	74.608867277	192.168.56.101	34.117.188.166	TCP	54	47298 → 443 [ACK] Seq=1 A
43	74.612282012	192.168.56.101	34.117.188.166	TLSv1.3	571	Client Hello
44	74.645178243	34.117.188.166	192.168.56.101	TCP	60	443 → 47308 [SYN, ACK] Se
45	74.645213166	192.168.56.101	34.117.188.166	TCP	54	47308 → 443 [ACK] Seq=1 A
46	74.659602981	192.168.56.101	34.117.188.166	TLSv1.3	571	Client Hello
47	74.713316097	34.107.221.82	192.168.56.101	TCP	60	80 → 45058 [SYN, ACK] Seq
48	74.713359911	192.168.56.101	34.107.221.82	TCP	54	45058 → 80 [ACK] Seq=1 Ac
49	74.713667188	192.168.56.101	34.107.221.82	HTTP	345	GET /canonical.html HTTP/
54	74.765806019	34.107.221.82	192.168.56.101	TCP	60	80 → 45058 [ACK] Seq=1 Ac
55	74.765806260	34.117.188.166	192.168.56.101	TCP	60	443 → 47308 [ACK] Seq=1 A

```
▸ Frame 1119: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface enp0s3, id 0
```

Figure (56) Filter traffic by Protocol (TCP)

3. Log Analysis

Action:

The packets generated by specific activities such as SSH, Ping, and file transfer were captured.

The logs were examined for the following details:

Source and destination addresses:

These can be observed in the "Source" and "Destination" columns.

Protocol types:

Displayed in the "Protocol" column.

Packet size:

Displayed in the "Length" column.

Timing:

The "Time" column was examined to monitor the timing of the packets.

Results:

SSH:

Encrypted traffic detected with SSH protocol.

Source and destination IP addresses are consistent with participating devices.

ICMP (Ping):

Display Echo request and response packets.

TCP:

Display large packets when transferring files using SCP.

The TCP protocol is responsible for ensuring secure transmission of packets.

Note: We have used 2 virtual machines, you may notice a slight difference in the names of the VM while loading Firewall Setup and Verification +SSH Configuration and Testing.